

Off-Policy Optimization and Tabular Q-Learning

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Recap



Recap

Policy Evaluation (PE):

- Estimating V^π , in an **unknown** discounted MDP, using data collected according to a **fixed** π
- Data could be from dataset (offline) or via interaction (online)
- TD update:

$$V(s) \leftarrow \begin{cases} V(s) + \alpha_t (r_t + \gamma V(s_{t+1}) - V(s)) & s = s_t \\ V(s) & \text{else.} \end{cases}$$

- If (i) π is exploratory enough, and (ii) $(\alpha_t)_t$ satisfies Robbins-Monro conditions:

$$V \rightarrow_{t \rightarrow \infty} V^\pi \quad \text{almost surely}$$



OPE/OPO

Two related problems:

- **Off-Policy Evaluation (OPE):** Estimate V^π of a **target policy** π using data collected according to some **behavior/logging policy** π_b
- **Off-Policy Optimization (OPO):** Find an **optimal policy** π^* using data collected according to some behavior policy π_b

This lecture: Two algorithms for OPO



Off-Policy Optimization

Off-Policy Optimization

Given: Data $\mathcal{D}$ collected under some policy π_b (not necessarily fixed).

Mathematically, $\mathcal{D} = \{(s_t, a_t, r_t), 1 \leq t \leq n\}$ where

$$a_t \sim \pi_b(\cdot|s_t), \quad r_t \sim R(s_t, a_t), \quad s_{t+1} \sim P(\cdot|s_t, a_t)$$

Goal: Find an optimal policy π^* , or a near-optimal one.



Action-Value Function (Q-Function)



Action-Value Function

The **action-value function** of policy π (or simply, **Q-value of π**) is a mapping $Q^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ defined as (Under the bounded reward assumption)

$$Q^\pi(s, a) := \mathbb{E}^\pi \left[\sum_{t=1}^{\infty} \gamma^{t-1} r(s_t, a_t) \mid s_0 = s, a_0 = a \right].$$

- Intuitively, $Q^\pi(s, a)$ measures the sum of future discounted rewards (in expectation) when the agent starts in s and takes action a in the first step (possibly $a \neq \pi(s)$), and then follows π afterwards.
- Again, recall that we assumed bounded rewards.
- We have

$$|Q^\pi(s, a)| \leq \frac{R_{\max}}{1 - \gamma}, \quad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}$$

- For all $s \in \mathcal{S}$, $Q^\pi(s, \pi(s)) = V^\pi(s)$.



Bellman Optimality Equation

Recall

$$V^* = \sup_{\pi \in \Pi^{\text{HR}}} V^\pi = \max_{\pi \in \Pi^{\text{SD}}} V^\pi$$

Q^* and V^* are related as

$$V^*(s) = \max_{a \in \mathcal{A}} Q^*(s, a)$$

Theorem

V^* and Q^* satisfy the *optimal Bellman equation*:

$$V^*(s) = \max_{a \in \mathcal{A}} \left(R(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x|s, a) V^*(x) \right), \quad s \in \mathcal{S}$$

$$Q^*(s, a) = R(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x|s, a) \max_{b \in \mathcal{A}} Q^*(x, b), \quad s \in \mathcal{S}, a \in \mathcal{A}$$



Optimality Theorems

A fundamental result in the theory of discounted MDPs:

Theorem

A stationary deterministic policy π is optimal if and only if it attains the maximum in the Bellman optimality equations: For all $s \in S$,

$$\pi(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \left(R(s, a) + \gamma \sum_{x \in S} P(x|s, a) V^*(x) \right) \quad \text{or equivalently,}$$

$$\pi(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \left(\underbrace{R(s, a) + \gamma \sum_{x \in S} P(x|s, a) \max_{b \in \mathcal{A}} Q^*(x, b)}_{=Q^*(s, a)} \right).$$

In short, the optimal policy is the greedy policy w.r.t. Q^* . Hence, enough to compute/learn Q^* .



Optimal Bellman Operator

The **optimal Bellman operator** is a mapping $\mathcal{T} : \mathbb{R}^{S \times A} \rightarrow \mathbb{R}^{S \times A}$, such that for any function $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$,

$$\mathcal{T}f(s, a) := R(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x|s, a) \max_{b \in \mathcal{A}} f(x, b), \quad s \in \mathcal{S}, a \in \mathcal{A}$$

$\mathcal{T}$ applies to (or *operates on*) a function defined on $\mathcal{S}$ and returns another function defined on $\mathcal{S}$.

- Q^* satisfies $\mathcal{T}Q^* = Q^*$.
- In words, Q^* is the *unique* fixed-point of the operator $\mathcal{T}^*$.



A Model-Based Method based on Certainty Equivalence



Known Model

When the model (MDP) is known, we simply solve the Bellman optimality equations using, e.g., VI or QVI:

QVI is quite similar to VI. It starts from Q_0 and iterates for $n \geq 1$:

$$Q_{n+1} = \mathcal{T}Q_n$$

I.e.,

$$Q_{n+1}(s, a) = R(s, a) + \gamma \sum_{x \in \mathcal{S}} P(x|s, a) \max_{b \in \mathcal{A}} Q_n(x, b)$$

$\Rightarrow Q_n$ converges to Q^* since $\mathcal{T}$ is contractive.



CE-OP0: Certainty Equivalence OPO

Model (MDP) is unknown, so one cannot solve the Bellman optimality equations.

Idea: Estimate the MDP using data and apply the [certainty equivalence principle](#).

- **Step 1:** Compute estimate $\hat{P}$ (of P) and $\hat{R}$ (of R)
- **Step 2:** Solve the Bellman optimality equations using $\hat{P}$ and $\hat{R}$



CE-OPO

Idea: Estimate the MDP using data and apply the **certainty equivalence principle**.

- **Step 1:** Compute estimate $\hat{P}$ (of P) and $\hat{R}$ (of R)
- **Step 2:** Solve the Bellman optimality equations using $\hat{P}$ and $\hat{R}$

We introduce visit counts for various triplets (s, a, s') .

Given a dataset $\mathcal{D} = \{(s_t, a_t, r_t), 1 \leq t \leq n\}$, define for any (s, a, s') ,

$$N(s, a, s') = \sum_{t=1}^{n-1} \mathbb{I}\{s_t = s, a_t = a, s_{t+1} = s'\}$$
$$N(s, a) = \sum_{s' \in \mathcal{S}} N(s, a, s')$$

A better choice in practice is

$$N(s, a) = \max \left\{ 1, \sum_{s' \in \mathcal{S}} N(s, a, s') \right\}$$



CE-OPO

Idea: Estimate the MDP using data and apply the **certainty equivalence principle**.

- **Step 1:** Compute estimate $\hat{P}$ (of P) and $\hat{R}$ (of R)
- **Step 2:** Solve the Bellman optimality equations using $\hat{P}$ and $\hat{R}$

Smoothed Estimator for P :

$$\hat{P}(s'|s, a) = \frac{N(s, a, s') + \alpha}{N(s, a) + \alpha S}$$

- S denotes the number of states.
- $\alpha \geq 0$ is an arbitrary choice controlling the level of smoothing.
- $\alpha = 0$ corresponds to Maximum Likelihood Estimator (unbiased).
- $\alpha = 1/S$ corresponds to Laplace Smoothed Estimator (biased, but the bias vanishes as $N(s, a)$ increases).
- If $\alpha = 0$, for $N(s, a) = 0$, define $\hat{P}(s'|s, a) = 1/S$.



CE-OPO

Idea: Estimate the MDP using data and apply the **certainty equivalence principle**.

- **Step 1:** Compute estimate $\hat{P}$ (of P) and $\hat{R}$ (of R)
- **Step 2:** Solve the Bellman optimality equations using $\hat{P}$ and $\hat{R}$

Smoothed Estimator for R :

$$\hat{R}(s, a) = \frac{\alpha + \sum_{t=1}^{n-1} r_t \mathbb{I}\{s_t = s, a_t = a\}}{\alpha + N(s, a)}$$

- $\alpha \geq 0$ is an arbitrary choice controlling the level of smoothing.
- $\alpha = 0$ corresponds to Maximum Likelihood Estimator (unbiased).



CE-OPO

Idea: Estimate the MDP using data and apply the **certainty equivalence principle**.

- **Step 1:** Compute estimate $\hat{P}$ (of P) and $\hat{R}$ (of R)
- **Step 2:** Solve the Bellman optimality equations using $\hat{P}$ and $\hat{R}$

Using $\hat{P}$ and $\hat{R}$, we can solve **empirical Bellman optimality equations**:

$$\hat{Q}^*(s, a) = \hat{R}(s, a) + \gamma \sum_{x \in \mathcal{S}} \hat{P}(x|s, a) \max_{b \in \mathcal{A}} \hat{Q}^*(x, b)$$

or

$$\hat{Q}^* = \hat{\mathcal{T}}\hat{Q}^*$$

$\hat{\mathcal{T}}$ is the empirical Bellman operator.

$\hat{Q}^* = \hat{\mathcal{T}}\hat{Q}^*$ can be solved using QVI.



CE-OP0: Certainty Equivalence OPO

CE-OP0: Certainty Equivalence OPO

- **input:** $\mathcal{D} = \{(s_t, a_t, r_t)\}_{1 \leq t \leq n}$, α (optional)
- Compute estimates $\hat{P}(s'|s, a)$ and $\hat{R}(s, a)$ for all (s, a, s')
- Find $\hat{\pi}^*$, the optimal policy in the empirical MDP $\hat{M} = (\mathcal{S}, \mathcal{A}, \hat{P}, \hat{R})$.
- **output:** $\hat{\pi}^*$

$\hat{M}$ could be solved using VI, PI, or QVI.



CE-OP0: Asymptotic Convergence

$$\hat{P}(s'|s, a) \xrightarrow{N(s,a) \rightarrow \infty} P(s'|s, a) \quad \text{almost surely}$$

$$\hat{R}(s, a) \xrightarrow{N(s,a) \rightarrow \infty} R(s, a) \quad \text{almost surely}$$

If π_b is exploratory enough in the sense that $N(s, a) \rightarrow_{n \rightarrow \infty} \infty$ for all (s, a) , then

$\hat{P}$ and $\hat{R}$ converge to P and R as $n \rightarrow \infty$. Thus, we can show

$$\hat{T} \xrightarrow{n \rightarrow \infty} T \quad \hat{Q}^* \xrightarrow{n \rightarrow \infty} Q^* \quad \text{almost surely}$$

which guarantees

$$\hat{\pi}^* \xrightarrow{n \rightarrow \infty} \pi^* \quad \text{almost surely}$$

Theorem

If all state-action pairs are visited *infinitely often* under π_b , then $\hat{Q}^*$ converges to Q^* almost surely:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \hat{Q}^* = Q^*\right) = 1$$



Strong guarantee, but only asymptotically (unfortunately).

CE-OP0: Pros and Cons

Disadvantages of the model-based solution:

- Often leads to large variance in the estimation of Q^*
- Computational complexity is $O(S^3)$, and space complexity is $O(S^2)$.
- May not be easily converted to an incremental procedure.



Model-Free Method for OPO: Tabular Q-Learning (and Friends)



Bellman Optimality Equations

Bellman optimality equations (using Q):

$$Q^*(s, a) = R(s, a) + \gamma \sum_{x \in S} P(x|s, a) \max_b Q^*(x, b)$$
$$\mathcal{T}Q^* = Q^*$$

Equivalently, Q^* is the root of functional $F(Q) = \mathcal{T}Q - Q$, namely the solution to the nonlinear system:

$$F(Q) = \mathcal{T}Q - Q = 0, \quad \text{where } Q \in \mathbb{R}^{S \times A}$$

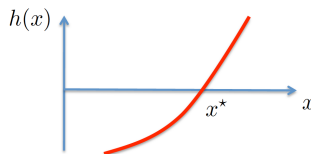
- **Known model:** Find the root of F using VI (or Q -iteration).
- **Unknown model:** We have only samples from R and P (as in TD).

We need a root finding method from noisy measurements.



Stochastic Approximation

Stochastic Approximation (SA) is a method to find the root of an increasing function from noisy measurements.



The setting:

- At the n -th iteration, you select x_n
- You get a noisy measurement $y_n = h(x_n) + e_n$
- e_n is a noise with zero-mean but may depend on the selected point x_n
- $\mathbb{E}[e_n \mid e_1, \dots, e_{n-1}] = 0$



Robbins-Monro Algorithm (1951)

SA was proposed by Robbins & Monro in (1951)

$$x_{n+1} = x_n - \alpha_n y_n = x_n - \alpha_n (h(x_n) + e_n), \quad n \geq 1$$

with $(\alpha_n)_n$ satisfying the **Robbins-Monro conditions**:

$$\alpha_n > 0, \quad \sum_{n=1}^{\infty} \alpha_n = \infty, \quad \text{and} \quad \sum_{n=1}^{\infty} \alpha_n^2 < \infty$$

Theorem

Under the following assumptions

- ❶ $\mathbb{E}[e_n \mid e_1, \dots, e_{n-1}] = 0$
- ❷ $\mathbb{E}[\|e_n\|^2 \mid e_1, \dots, e_{n-1}] \leq K(1 + \|x_n\|^2)$, *almost surely for some K*
- ❸ h is Lipschitz
- ❹ $\sup_n \|x_n\| < \infty$, *almost surely*

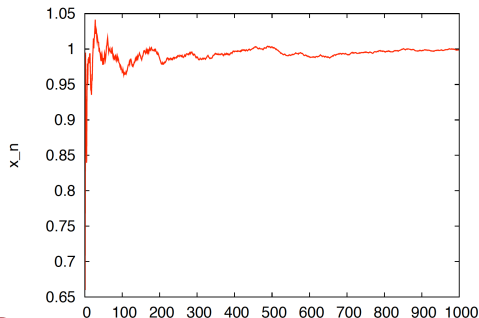
$$\lim_{n \rightarrow \infty} x_n = x^* \quad \text{almost surely.}$$



Example

Solving $h(x) = x^2 - 1 = 0$ through noisy samples from h using SA, with $x_0 = 1$ and $\alpha_n = 1/n$:

$$x_{n+1} = x_n - \frac{y_n}{n}$$



SA for $F(Q) = \mathcal{T}Q - Q$

We apply SA to $F(Q) = \mathcal{T}Q - Q$.

- Consider a sample (s_t, a_t, r_t, s_{t+1}) .
- We show that $Y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)$, **conditioned on (s_t, a_t) and Q** is an unbiased sample from $F(Q)(s_t, a_t)$.

$$\begin{aligned}
 \mathbb{E}[Y_t | Q, s_t, a_t] &= \mathbb{E}\left[r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \middle| Q, s_t, a_t\right] \\
 &= \underbrace{\mathbb{E}\left[r_t \middle| Q, s_t, a_t\right]}_{=R(s_t, a_t)} + \gamma \mathbb{E}\left[\max_{a'} Q(s_{t+1}, a') \middle| Q, s_t, a_t\right] - Q(s_t, a_t) \\
 &= R(s_t, a_t) + \gamma \sum_{x \in \mathcal{S}} P(x | s_t, a_t) \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \\
 &= \mathcal{T}Q(s_t, a_t) - Q(s_t, a_t) = F(Q)(s_t, a_t)
 \end{aligned}$$

Hence, $\mathbb{E}[Y_t | \mathcal{H}_{t-1}] = F(Q)(s_t, a_t)$.

- The other technical conditions of SA can be verified. (Technical and tedious, so omitted here.)



SA for $F(Q) = \mathcal{T}Q - Q$

Application of SA to $F(Q) = \mathcal{T}Q - Q$:

The **Q-Learning (QL)** update rule:

$$\underbrace{Q(s_t, a_t)}_{\text{new value}} \leftarrow \underbrace{Q(s_t, a_t)}_{\text{old value}} + \underbrace{\alpha_t \left(r_t + \gamma \max_{b \in \mathcal{A}} Q(s_{t+1}, b) - Q(s_t, a_t) \right)}_{\text{correction}}$$

And $Q(s, a)$ unchanged if $(s, a) \neq (s_t, a_t)$.



QL: Learning Rate

To guarantee convergence, learning rates $(\alpha_t)_{t \geq 1}$ must satisfy the *Robbins-Monro conditions*:

$$\alpha_t > 0, \quad \sum_{t=1}^{\infty} \alpha_t = \infty, \quad \sum_{t=1}^{\infty} \alpha_t^2 < \infty$$

(I.e., a positive sequence that is *square-summable-but-not-summable*.)

Examples:

- $\alpha_t = \frac{1}{t+1}$,
- $\alpha_t = \frac{c}{t^a}$ for $a \in (\frac{1}{2}, 1]$
- $\alpha_t = \alpha_t(s, a) = \frac{1}{N_t(s, a) + 1}$, where $N_t(s, a)$ is the number of times (s, a) is sampled in the first $t - 1$ rounds —i.e., learning rate can be personalized to (s, a) , assuming that Robbins-Monro conditions could be met.



QL

input: $\mathcal{D} = \{(s_t, a_t, r_t)\}_{1 \leq t \leq n}, (\alpha_t)_{t \geq 1}$

initialization: Select Q_1 arbitrarily

for $t = 1, \dots, n - 1$:

- $\delta_t = r_t + \gamma \max_{b \in \mathcal{A}} Q_t(s_{t+1}, b) - Q_t(s_t, a_t)$
- Update:

$$Q_{t+1}(s, a) = \begin{cases} Q_t(s, a) + \alpha_t \delta_t & (s, a) = (s_t, a_t) \\ Q_t(s, a) & \text{else.} \end{cases}$$

output: Greedy policy w.r.t. Q_n

- Q_n is an estimate of Q^* , giving an estimate $\widehat{\pi^*}$ of the optimal policy:

$$\widehat{\pi^*}(s) \in \operatorname{argmax}_{a \in \mathcal{A}} Q_n(s, a)$$



QL: Asymptotic Convergence

Theorem

If all state-action pairs are visited *infinitely often in $\mathcal{D}$* and $(\alpha_t)_{t \geq 1}$ satisfies the Robbins-Monro conditions, then Q_t converges to the true value function Q^* almost surely:

$$\mathbb{P} \left(\forall s \in \mathcal{S}, a \in \mathcal{A}, \lim_{t \rightarrow \infty} Q_t(s, a) = Q^*(s, a) \right) = 1$$

In other words, if π_b (used to collect $\mathcal{D}$) is exploratory enough, Q_t converges to Q^* , in the following sense:

$$\mathbb{P} \left(\exists \mathcal{D}, \exists (s, a) : \lim_{t \rightarrow \infty} Q_t(s, a; \mathcal{D}) \neq Q^*(s, a) \right) = 0$$

I.e., datasets for which $Q_\infty \neq Q^*$ will occur with probability 0.



On Behavior Policy

- (Asymptotic) convergence requires that state-action pairs are visited infinitely often.
- The behavior policy π_b could change during the learning, as long as it is kept exploratory enough.
 - E.g., ε -greedy policy (for some $\varepsilon > 0$)

$$\pi_{\varepsilon\text{-greedy}}(s) = \begin{cases} \operatorname{argmax}_a Q_t(s, a) & \text{w.p. } 1 - \varepsilon \\ \text{sample uniformly at random from } \mathcal{A} & \text{w.p. } \varepsilon \end{cases}$$

Note that $\pi_{\varepsilon\text{-greedy}}(s)$ is non-stationary.

- E.g., Boltzmann's policy (a.k.a. softmax):

$$\text{at state } s, \text{ select action } a \in \mathcal{A} \text{ w.p. } \frac{e^{\eta Q_t(s, a)}}{\sum_{b \in \mathcal{A}} e^{\eta Q_t(s, b)}}$$

where $\eta > 0$ is a parameter controlling exploration.

- Incremental QL (cf. the very last slides)



QL: Advantages

- QL is **model-free**: It does not require to estimate a model of the MDP, and only relies on collected experience.
- QL can be incremental (unlike the model-based methods).
- Space complexity is $O(SA)$ and computational complexity, per round, is $O(A)$. Much cheaper than the model-based method.



QL: Non-Asymptotic Convergence

- Asymptotic convergence results often do not tell us much information about the speeds of convergence.
- We are interested in knowing what happens with small datasets. So we study the non-asymptotic convergence.

Sample complexity for OPO

Given $\delta \in (0, 1)$ and $\varepsilon > 0$, define the **PAC off-policy sample complexity** as the number $SC(\varepsilon, \delta)$ of samples from the MDP such that for all $n \geq SC(\varepsilon, \delta)$,

$$\|Q^* - Q_n\|_\infty \leq \varepsilon, \quad \text{with probability } \geq 1 - \delta$$



Two Definitions

Two notions arising in sample complexity of OPO:

Cover Time t_{cover} . Given $t_1 > 0$, let $t_2 > t_1$ denote the first time step such that all (s, a) pairs are visited at least once with probability at least $\frac{1}{2}$. Then, $t_{\text{cover}} = t_2 - t_1$ defines the **cover time** of M .

- $t_{\text{cover}} \geq SA$.
- A quantity related to π_b .

Effective Horizon. Given $\varepsilon > 0$, the **effective horizon** is

$$H_{\text{eff}} := \frac{-1}{1 - \gamma} \log(\varepsilon(1 - \gamma))$$

- Truncating ∞ -horizon to H_{eff} would bring at most ε error to V^* .



QL: Non-Asymptotic Convergence

Theorem (Even-dar & Mansour (2003))

Let $\delta \in (0, 1)$ and $\varepsilon \in (0, \frac{1}{1-\gamma}]$, and assume that n satisfies:

$$n \geq c \cdot \frac{[t_{\text{cover}}]^{H_{\text{eff}}}}{\varepsilon^2(1-\gamma)^4} \log\left(\frac{SA n}{\delta}\right) \log\left(\frac{SA}{\varepsilon \delta(1-\gamma)^2}\right)$$

where c is a universal constant. Then, QL with $\alpha_t(s, a) = \frac{1}{N_t(s, a) + 1}$ satisfies:

$$\|Q^* - Q_n\|_{\infty} \leq \varepsilon, \quad \text{with probability } \geq 1 - \delta.$$

Essentially, it establishes a sample complexity for QL proportional to

$$\tilde{O}\left(\frac{[t_{\text{cover}}]^{H_{\text{eff}}}}{\varepsilon^2(1-\gamma)^4}\right)$$

where $\tilde{O}(\cdot)$ hides poly-log terms.



QL: Non-Asymptotic Convergence

Theorem (Li et al. (2020))

Let $\delta \in (0, 1)$ and $\varepsilon \in (0, \frac{1}{1-\gamma}]$, under QL one has:

$$\|Q^* - Q_n\|_\infty \leq \varepsilon, \quad \text{with probability } \geq 1 - \delta.$$

provided that

$$n \geq c \cdot \frac{t_{\text{cover}}}{\varepsilon^2(1-\gamma)^5} \log^2\left(\frac{SAn}{\delta}\right) \log\left(\frac{1}{\varepsilon(1-\gamma)^2}\right)$$
$$\alpha_t = \frac{c'}{\log(SAn/\delta)} \min\left(\frac{(1-\gamma)^4 \varepsilon^2}{\gamma^2}, 1\right)$$

where c, c' are universal constants.

Essentially, it establishes a sample complexity for QL proportional to

$$\tilde{O}\left(\frac{t_{\text{cover}}}{\varepsilon^2(1-\gamma)^5}\right)$$



QL: Overestimation Bias

QL could exhibit weak empirical performance due overestimation bias.

- Overestimation bias stems from the term

$$\max_{b \in \mathcal{A}} Q_t(s_{t+1}, b)$$

to approximate $\max_{b \in \mathcal{A}} Q^*(s_{t+1}, b)$ in the update equation of QL.

- It is one major reason behind slow convergence of QL in practice.

Could we update Q_t in a wiser way?

Idea: $\max_{b \in \mathcal{A}} Q^*(s_{t+1}, b)$ is related to the classical problem of **Estimating the Maximum Expected Value**. So let's use a wiser such estimate.



Estimating the Maximum Expected Value

Consider r.v.'s $X_1, \dots, X_m$ with $\mathbb{E}[X_i] = \mu_i$.

- We wish to estimate $\mu_\star = \max_i \mathbb{E}[X_i]$.
- Distributions of $X_i, \dots, X_m$ unknown.
- We have a set S_i of i.i.d. samples from each X_i .

Maximum Estimator (ME): We construct $\hat{\mu}_i := \hat{\mu}_i(S_i) = \frac{1}{|S_i|} \sum_{x \in S_i} x$, and set

$$\hat{\mu}_\star^{\text{ME}} := \max_i \hat{\mu}_i.$$

$\hat{\mu}_\star^{\text{ME}}$ is **positively biased** since: $\mathbb{E}[\hat{\mu}_\star^{\text{ME}}] = \mathbb{E}[\max_i \hat{\mu}_i] \geq \max_i \mathbb{E}[\hat{\mu}_i] = \max_i \mu_i = \mu_\star$

Double Estimator (DE): Randomly partition each sample set as $S_i = S_i^A \cup S_i^B$.

$$\bar{i} \in \operatorname{argmax}_i \hat{\mu}_i(S_i^A) \quad \text{Then} \quad \hat{\mu}_\star^{\text{DE}} := \hat{\mu}_{\bar{i}}(S_{\bar{i}}^B).$$

It can be shown that $\hat{\mu}_\star^{\text{DE}}$ is **negatively biased**.



Combining Double Estimator with QL

The Double Estimator could be incorporated into QL:

- Let's maintain two estimates of Q-values Q^A and Q^B , each updates using half of the samples from $\mathcal{D}$:
- Update for Q^A

$$Q_{t+1}^A(s, a) = \begin{cases} Q_t^A(s, a) + \alpha_t \left(r_t + \gamma Q_t^B(s_{t+1}, \bar{a}) - Q_t^A(s, a) \right) & (s, a) = (s_t, a_t) \\ Q_t^A(s, a) & \text{else.} \end{cases}$$

with $\bar{a} = \operatorname{argmax}_b Q_t^A(s_{t+1}, b)$.

- A similar update will be made for Q^B

The corresponding algorithm is called `Double QL` (van Hasselt, 2010).



Double QL

input: $\mathcal{D} = \{(s_t, a_t, r_t)\}_{1 \leq t \leq n}, (\alpha_t)_{t \geq 1}$

initialization: Select Q_1^A, Q_1^B arbitrarily

for $t = 1, \dots, n - 1$:

- Set update-A = True w.p. 0.5
- **if** update-A:
 - $\bar{a} = \operatorname{argmax}_a Q_t^A(s_{t+1}, a)$
 - $\delta_t = r_t + \gamma Q_t^B(s_{t+1}, \bar{a}) - Q_t^A(s_t, a_t)$
 - Update: $Q_{t+1}^A(s, a) = \begin{cases} Q_t^A(s, a) + \alpha_t \delta_t & (s, a) = (s_t, a_t) \\ Q_t^A(s, a) & \text{else.} \end{cases}$
- **else:**
 - $\bar{a} = \operatorname{argmax}_a Q_t^B(s_{t+1}, a)$
 - $\delta_t = r_t + \gamma Q_t^A(s_{t+1}, \bar{a}) - Q_t^B(s_t, a_t)$
 - Update: $Q_{t+1}^B(s, a) = \begin{cases} Q_t^B(s, a) + \alpha_t \delta_t & (s, a) = (s_t, a_t) \\ Q_t^B(s, a) & \text{else.} \end{cases}$

output: Policy greedy w.r.t. $Q_n^A + Q_n^B$

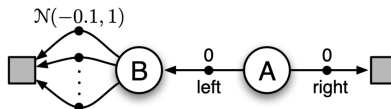
- Q_n is an estimate of Q^* , giving an estimate $\widehat{\pi}^*$ of the optimal policy:

$$\widehat{\pi}^*(s) \in \operatorname{argmax}_{a \in \mathcal{A}} Q_n^A(s, a) + Q_n^B(s, a)$$



Double QL vs. QL

Double QL vs. QL in a simple MDP (Source: Sutton & Barto):



- 4 states: A , B and two terminal states denoted by $\square$
- At A : Two actions ('left' and 'right'), each with $r = 0$
- At B : Multiple actions, each with $r \sim \mathcal{N}(-0.1, 1)$
- $\implies \pi^*(A) = \text{right}$

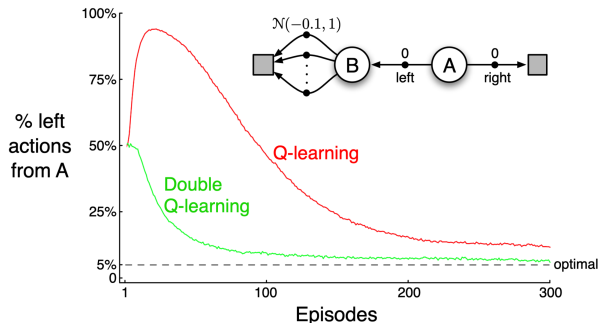
However, OPO methods may choose 'left' since **maximization bias** making B appear to have a positive value.



Double QL vs. QL

Double QL vs. QL in a simple MDP (Source: Sutton & Barto):

Averaged over 10000 runs. π_b is ϵ -greedy with $\epsilon = 0.1$.



- QL initially learns 'left' much more often than 'right'
- In contrast, Double QL is less affected by maximization bias.



Off-Policy vs. Offline

Off-Policy Learning/Optimization $\neq$ Offline RL

- In offline RL, the goal is to learn an optimal policy (or a near-optimal one) from a dataset – *we're offline; no further exploration.*
- Offline RL $\subset$ OPO
- Note that OPO could take place in an online fashion (*but behavior must be generated off-the-target-policy*)



Historical Account

- Christopher Watkins presented QL in 1989 in his PhD thesis.
- In 1994, Tsitsiklis established the almost sure convergence of QL by showing its relation to SA. See the paper for a detailed proof of asymptotic convergence guarantee and verification of SA conditions (Tsitsiklis, 1994).
- Non-asymptotic convergence of QL was done in (Even-dar & Mansour, 2003). State-of-the-art is (Li et al., 2020).
- Double QL is presented in (van Hasselt, 2010).
- Research on improved sample complexity of QL as well as improved variants is ongoing.



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